

Alejandro Sánchez Yalí

AI Engineer



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SUMMARY

AI Engineer and mathematician with 8+ years building production software, 3+ years shipping LLM-powered systems end to end and two years (2018–2020) building blockchain platforms. Designs and operates agent backends: multi-turn systems with tool calling, RAG, state management and human handoff, multi-provider LLM gateways with fallback, and the reliability work around them: timeouts, retries, graceful degradation and tracing. M.Sc. in Mathematics.

TECHNICAL SKILLS

- **AI & LLMs:** LangChain, LangGraph, LiteLLM, RAG, MCP, OpenAI API, Whisper, ElevenLabs, agentic development with Claude Code
- **ML & Mathematics:** PyTorch, TensorFlow, Scikit-learn, JAX, Flax, NumPy, Pandas, linear algebra, probability, statistical inference
- **Languages:** Python (8+ years), TypeScript (6+ years), JavaScript (6+ years), SQL, Rust (personal projects)
- **Backend:** FastAPI, Django, Celery, Flask, NestJS, Node.js, RESTful APIs, microservices, event-driven design
- **Data & Infrastructure:** PostgreSQL, Redis, MongoDB, Docker, AWS, GCP, GitHub Actions, CI/CD
- **Architecture:** Clean architecture, DDD, SOLID, modular monolith, screaming architecture

EXPERIENCE

AI Specialist

Lapzo

Aug 2025 – Sep 2026

Remote, Mexico

- Built an HR agent (LangGraph, FastAPI, PostgreSQL) that answers company-policy questions with citations and executes HR requests with human confirmation, with per-tenant isolation; shipped to QA and staging with a runbook and a handover
- Migrated the course-vectorization system to a dedicated VM and moved the embeddings pipeline from n8n into the platform; hardened the LLM course assistant (scoped context, consistent recommendations)
- Designed and built a Digital Professor that holds real-time conversations with learners and explains course content through interactive slides (LLM services, ElevenLabs voice, real-time layer)
- Built the audit system for courses and learning paths and an AI-powered educational content generator with LangChain, LangGraph and n8n, with Celery workers running the pipelines in the background

Independent Software & AI Consultant

Freelance

Aug 2024 – Present

Remote

- Lead engineer of Aluna (AI recruitment platform, see below) and builder of a multi-tenant AI customer service agent for CREARIA (Colombia) on WhatsApp Cloud API, LiteLLM, RAG and MCP with automatic human escalation
- Led technical design sessions with clients, translating business requirements into system architectures and naming the trade-off behind each technology decision
- Use Claude Code daily on live client projects, steering and critically reviewing generated output to ship end-to-end features without eroding architectural boundaries
- Designed database architecture for a client's workflow automation platform and served as code reviewer for their development team

Full Stack Engineer

Monadical

Apr 2021 – Aug 2024

Remote, Canada/US

- Built and integrated AI services for information retrieval with large language models (OpenAI, LangChain, Whisper) in production web applications
- Participated in system design and architectural decisions, evaluating trade-offs across performance, scalability and maintainability
- Built and scaled microservices for AI/ML and data-intensive applications; designed ETL pipelines and custom Django ORM extensions
- Did code reviews on a distributed team across Canada, the US and Latin America

Full Stack Engineer

BCFort

Aug 2018 – Oct 2020

Medellín, Colombia

- Designed system architecture for blockchain platforms: smart contract patterns, data models and service boundaries
- Developed blockchain and analytics platforms with Python (Django), Node.js, React and Web3.js on AWS, GCP and IBM Cloud

Professor of Mathematics & Machine Learning

Universidad de Antioquia

Jan 2010 – Oct 2021

Medellín, Colombia

- Taught Machine Learning, Data Science and Mathematics with Python, NumPy, Pandas, TensorFlow, PyTorch and Scikit-learn
- Developed the AI/ML elective curriculum for the Mathematics Institute and mentored student research in ML and data analysis

FEATURED PROJECTS

- **Plixiq – AI Customer Support on WhatsApp**  – *Python, FastAPI, LiteLLM, PostgreSQL, Redis, Next.js, WebSocket*
 - Sole architect and engineer of a multi-tenant SaaS where AI agents answer routine customer inquiries, book appointments and escalate the rest to human operators with full conversation context
 - Modular monolith with 14 bounded contexts enforced in CI; multi-provider LLM layer (Groq default, OpenAI fallback) so switching providers is a config change
- **Aluna – AI Recruitment Platform for Staffing Agencies**  – *Next.js 15, FastAPI, LiteLLM, PostgreSQL, Inngest, Claude, Gemini, WhatsApp Cloud API*
 - Lead engineer of an AI recruitment platform for staffing agencies, built in 3 months – 336 source files across 4 packages, 26 tables and 11 external providers – split into a Next.js 15 application owning all domain state and a stateless FastAPI service owning model calls and the WhatsApp channel
 - Durable Inngest analysis pipeline (9 checkpointed steps, 3 retries) where deduplication runs before quota enforcement so a re-analysed CV reuses its cached score instead of paying for another model call; one screening engine serves WhatsApp, web chat and email
- **Digital Human – Interactive AI Avatar**  – *React, Three.js, React Three Fiber, OpenAI GPT, Whisper, ElevenLabs*
 - Open-source conversational avatar with 464 stars and 110 forks: real-time speech-to-text, generation, voice synthesis and lip synchronisation
 - The engineering problem was latency budgeting across four services, since a human notices any pause in a conversation

OPEN SOURCE

- **talking-avatar-with-ai**  – 464 stars, 110 forks. Open-source conversational AI avatar: OpenAI GPT, Whisper, ElevenLabs, Rhubarb lip-sync, Three.js
- **Monadical-SAS/Morpheus**  – 19 pull requests. Open-source AI image generation platform: FastAPI, React, Stable Diffusion
- 493 GitHub stars across own open-source projects – top 4.8% of ranked GitHub profiles (GitRanks)

CONFERENCE TALKS

- **Scaling Learning Models: Parallelism Strategies in JAX with Flax**  – PyCon Colombia, 2024

TECHNICAL WRITING

- **Pre-training: the Idea Behind Every LLM** (2026)
- **Neural Networks as DAGs of Parameterized Computational Programs** (2024)
- **Revolutionizing Animation: Building Digital Humans with LLMs** (2024)

EDUCATION

M.Sc. in Mathematics

Universidad de Antioquia

Aug 2010 - Sep 2013

Medellín, Colombia

B.Sc. in Teaching Mathematics and Physics

Universidad de Antioquia

Jan 2004 - Oct 2009

Medellín, Colombia